

## PINAR DEL RIO, Cuba

WMO: 783150

Lat: 22.417N

Lon: 83.683W

Elev: 121

StdP: 14.63

Time Zone: -5.00 (CUB)

Period: 82-15

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |      |      |      |      |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|------|------|------|------|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |      |      | 99%  |      |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR   | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e)  | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | 55.4       | 58.1 | 48.2                          | 50.1 | 62.4 | 50.5 | 54.7 | 64.3 | 18.7                  | 73.6 | 17.3 | 73.8 | 0.8                   | 40   | 0.243 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 8             | 14.1                   | 90.6            | 77.2 | 89.5 | 77.0 | 88.6 | 76.9 | 80.4                | 86.5 | 79.7 | 86.0 | 79.0 | 85.5 | 4.6                  | 90   |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 78.7                            | 149.8 | 84.2 | 77.9 | 145.5 | 83.4 | 77.2 | 142.1 | 82.7 | 44.0          | 86.7 | 43.2 | 85.9 | 42.5 | 85.6 | 86.2              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |     |                    |     | n-Year Return Period Values of Extreme Temperature |     |            |     |            |     |            |     |
|-------------------|------|------|----------------------------|-----|--------------------|-----|----------------------------------------------------|-----|------------|-----|------------|-----|------------|-----|
| 1%                | 2.5% | 5%   | Mean                       |     | Standard Deviation |     | n=5 years                                          |     | n=10 years |     | n=20 years |     | n=50 years |     |
|                   |      |      | Min                        | Max | Min                | Max | Min                                                | Max | Min        | Max | Min        | Max | Min        | Max |
| (a)               | (b)  | (c)  | (e)                        | (f) | (g)                | (h) | (i)                                                | (j) | (k)        | (l) | (m)        | (n) | (o)        | (p) |
| 16.2              | 13.9 | 11.7 | N/A                        | N/A | N/A                | N/A | N/A                                                | N/A | N/A        | N/A | N/A        | N/A | N/A        | N/A |
|                   |      |      | N/A                        | N/A | N/A                | N/A | N/A                                                | N/A | N/A        | N/A | N/A        | N/A | N/A        | N/A |
|                   |      |      | DB                         |     |                    |     |                                                    |     |            |     |            |     |            |     |
|                   |      |      | WB                         |     |                    |     |                                                    |     |            |     |            |     |            |     |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|------------------------------------------------------------------------|-------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                        |             | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 76.8        | 71.1  | 72.4  | 73.7  | 77.0  | 79.2  | 80.5  | 81.1  | 81.5  | 80.4  | 78.0  | 75.2  | 71.7  |
|                                                                        | DBStd       | 4.73        | 3.76  | 3.89  | 3.99  | 2.86  | 1.98  | 1.92  | 1.84  | 1.67  | 1.64  | 2.48  | 3.40  | 4.04  |
|                                                                        | HDD50       | 0           | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
|                                                                        | HDD65       | 15          | 5     | 3     | 1     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 6     |
|                                                                        | CDD50       | 9798        | 653   | 628   | 734   | 810   | 906   | 916   | 965   | 976   | 911   | 868   | 757   | 674   |
|                                                                        | CDD65       | 4337        | 193   | 211   | 270   | 360   | 441   | 466   | 500   | 511   | 461   | 403   | 307   | 214   |
|                                                                        | CDH74       | 32983       | 904   | 1207  | 1860  | 2825  | 3597  | 4105  | 4483  | 4614  | 3683  | 2867  | 1822  | 1016  |
|                                                                        | CDH80       | 9878        | 95    | 212   | 422   | 852   | 1231  | 1346  | 1570  | 1688  | 1171  | 791   | 381   | 119   |
| (14) Wind                                                              | WSAvg       | 3.3         | 3.6   | 3.8   | 4.3   | 3.9   | 3.6   | 3.0   | 2.6   | 2.4   | 2.4   | 3.0   | 3.5   | 3.6   |
| (15) Precipitation                                                     | PrecAvg     | 55.0        | 2.5   | 1.8   | 2.2   | 3.1   | 5.8   | 8.4   | 5.5   | 6.4   | 8.3   | 6.3   | 2.2   | 1.8   |
|                                                                        | PrecMax     | 76.6        | 9.0   | 3.3   | 5.3   | 8.7   | 14.0  | 22.8  | 11.1  | 11.9  | 18.0  | 13.1  | 7.0   | 6.3   |
|                                                                        | PrecMin     | 40.3        | 0.2   | 0.5   | 0.1   | 0.7   | 1.6   | 3.0   | 3.2   | 4.0   | 3.4   | 1.8   | 0.1   | 0.1   |
|                                                                        | PrecStd     | 9.8         | 2.5   | 0.8   | 1.4   | 1.9   | 3.1   | 5.4   | 1.9   | 2.5   | 3.3   | 3.3   | 1.6   | 1.7   |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB          | 84.7  | 86.0  | 87.7  | 89.3  | 90.9  | 90.8  | 91.5  | 92.6  | 90.6  | 89.1  | 87.2  | 84.4  |
|                                                                        |             | MCWB        | 72.1  | 71.9  | 74.1  | 73.3  | 74.0  | 77.6  | 77.7  | 77.4  | 77.4  | 77.0  | 75.2  | 72.3  |
|                                                                        | 2%          | DB          | 82.6  | 84.1  | 85.6  | 87.6  | 89.1  | 89.2  | 89.9  | 90.5  | 89.1  | 87.3  | 85.5  | 82.8  |
|                                                                        |             | MCWB        | 71.6  | 71.5  | 73.1  | 73.2  | 74.0  | 77.6  | 77.6  | 78.1  | 77.6  | 76.5  | 74.7  | 71.9  |
|                                                                        | 5%          | DB          | 81.2  | 82.5  | 84.2  | 86.5  | 87.7  | 88.0  | 88.7  | 89.2  | 87.8  | 86.1  | 84.3  | 81.4  |
|                                                                        |             | MCWB        | 70.9  | 71.3  | 72.4  | 73.0  | 74.2  | 77.7  | 77.6  | 77.9  | 77.7  | 76.4  | 74.5  | 71.4  |
|                                                                        | 10%         | DB          | 79.5  | 80.8  | 82.6  | 84.9  | 86.2  | 86.7  | 87.4  | 87.9  | 86.3  | 84.7  | 82.7  | 79.7  |
|                                                                        |             | MCWB        | 70.2  | 70.9  | 71.7  | 72.8  | 74.0  | 77.6  | 77.6  | 78.1  | 77.6  | 75.6  | 74.1  | 70.8  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB          | 75.4  | 75.4  | 77.2  | 77.4  | 79.3  | 81.7  | 80.9  | 81.6  | 80.1  | 79.6  | 78.0  | 75.4  |
|                                                                        |             | MCDB        | 80.9  | 81.6  | 84.4  | 84.2  | 85.8  | 86.2  | 87.3  | 88.3  | 86.1  | 85.2  | 83.2  | 80.8  |
|                                                                        | 2%          | WB          | 73.8  | 74.3  | 75.5  | 76.3  | 78.2  | 80.1  | 79.8  | 80.4  | 79.3  | 78.3  | 76.6  | 73.8  |
|                                                                        |             | MCDB        | 79.1  | 80.4  | 82.5  | 83.4  | 84.7  | 85.5  | 86.4  | 87.0  | 85.6  | 84.0  | 82.4  | 79.4  |
|                                                                        | 5%          | WB          | 72.7  | 73.2  | 74.4  | 75.4  | 77.1  | 79.2  | 79.0  | 79.4  | 78.8  | 77.7  | 75.7  | 72.9  |
|                                                                        |             | MCDB        | 78.2  | 79.2  | 81.2  | 82.5  | 83.6  | 85.0  | 85.8  | 86.3  | 85.3  | 83.3  | 81.7  | 78.4  |
|                                                                        | 10%         | WB          | 71.6  | 72.1  | 73.3  | 74.5  | 76.0  | 78.4  | 78.3  | 78.6  | 78.2  | 76.9  | 74.9  | 71.9  |
|                                                                        |             | MCDB        | 77.3  | 78.0  | 80.1  | 81.5  | 82.2  | 84.4  | 85.0  | 85.5  | 84.4  | 82.4  | 80.8  | 77.4  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR        | 16.3  | 16.9  | 17.0  | 16.4  | 14.5  | 12.6  | 13.5  | 14.1  | 13.0  | 13.4  | 14.2  | 15.4  |
|                                                                        |             | MCDBR       | 17.2  | 17.7  | 17.5  | 17.0  | 16.5  | 14.5  | 15.1  | 15.5  | 14.9  | 14.9  | 14.8  | 15.9  |
|                                                                        | 5% WB       | MCWBR       | 8.0   | 7.7   | 7.2   | 6.0   | 5.5   | 5.6   | 5.6   | 6.0   | 6.0   | 5.9   | 6.0   | 6.9   |
|                                                                        |             | MCWBR       | 15.0  | 15.8  | 16.4  | 15.4  | 13.7  | 12.3  | 13.7  | 14.1  | 13.2  | 12.6  | 13.4  | 14.1  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |             | 0.365 | 0.370 | 0.375 | 0.406 | 0.431 | 0.450 | 0.466 | 0.433 | 0.417 | 0.401 | 0.373 | 0.367 |
|                                                                        |             | taud        | 2.518 | 2.511 | 2.478 | 2.389 | 2.365 | 2.359 | 2.303 | 2.423 | 2.453 | 2.489 | 2.514 | 2.539 |
|                                                                        | Ebn at Noon |             | 286   | 292   | 294   | 285   | 275   | 268   | 264   | 273   | 277   | 279   | 281   | 282   |
|                                                                        |             | Edn at Noon | 31    | 33    | 35    | 39    | 40    | 40    | 42    | 37    | 36    | 33    | 30    | 29    |
| (44) All-Sky Solar Radiation                                           | RadAvg      |             | 1267  | 1534  | 1831  | 2028  | 1946  | 1718  | 1901  | 1814  | 1626  | 1480  | 1322  | 1214  |
|                                                                        | RadStd      |             | 95    | 103   | 71    | 72    | 165   | 129   | 90    | 64    | 104   | 118   | 69    | 90    |

## Historical Trends

|                             | DBAvg | Heating |     |        | Cooling |       |       | Degree-Days |       |       |       |
|-----------------------------|-------|---------|-----|--------|---------|-------|-------|-------------|-------|-------|-------|
|                             |       | 99% DB  |     | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                             |       | (d)     | (e) | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |
| (46) Station Only           | N/A   | N/A     | N/A | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (2 neighbors) | N/A   | N/A     | N/A | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page